

Math 113 – Abstract Algebra

Comprehensive exercise solutions

Fraleigh, *A First Course in Abstract Algebra* (7th ed.) ■ Fall 2026 ■ Keshav Ramamurthy

Adding a solution

To record a solved exercise, type under its chapter banner:

```
\exercise{2.14} \begin{solution} ... \end{solution}
```

Restate the problem first with `\begin{problem}[title]... \end{problem}`; structure proofs with `claim`, `idea`, `proofsketch`.

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Chapter 0 Sets and Relations

Fraleigh: sets, relations, equivalence relations, partitions, functions

Chapter 1 Groups and Subgroups

Fraleigh: examples, binary operations, isomorphism, groups, subgroups, cyclic groups

Chapter 2 Permutations, Cosets, and Direct Products

Fraleigh: cycle notation, Lagrange's theorem, direct products

Chapter 3 Homomorphisms and Factor Groups

Fraleigh: homomorphisms, normal subgroups, factor groups, the isomorphism theorems

Chapter 4 Rings and Fields

Fraleigh: rings, fields, integral domains, units and zero divisors

Chapter 5 Ideals and Factor Rings

Fraleigh: ideals, factor rings, prime and maximal ideals, PIDs

Chapter 6 Extension Fields

Fraleigh: vector spaces over fields, algebraic extensions, finite fields

Chapter 7 Advanced Group Theory

Fraleigh: Sylow theorems, group actions, simple groups, composition series

Chapter 9 Factorization

Fraleigh: UFDs, Euclidean domains, polynomial factorization

Chapter 10 Automorphisms and Galois Theory

Fraleigh: automorphisms, fixed fields, the Galois correspondence